Project Requirements

In the project you will form groups of up to 4 students. You will

1. Construct or download an interesting multimodal dataset. Multimodal means the dataset should include at least one other type of data (text, image, network, sensor etc.) in addition to numerical/categorical data. You may use APIs, government data, financial data, or any other data sources of interest. *For example, you could combine a database of all of Donald Trump's tweets since 2017 with financial data on the US dollar from FRED-MD.*

2. Clean your data: deal with missing values, extract features, and combine different data sources.

3. Extract insights from your data using at least one unsupervised learning method, such as clustering, sentiment analysis, or dimension reduction. *For example, you could analyze the sentiment of Trump's tweets over time, and describe how the sentiment changed between his first and second presidency.*

4. Use multiple supervised learning methods and compare their performance on at least one prediction task. Use appropriate out-of-sample metrics. Always report the performance of at least one very simple model as a baseline, such as a version of linear or logistic regression. Provide some explanation and intuition on why certain methods perform well. *For example, you could use aggregated features from Trump tweets to predict monthly (or daily) changes in the US dollar index.*

You are encouraged to use both methods that we learned in class, as well as methods that go in some way beyond what we learned in class (e.g. different neural network architectures).

You are encouraged to choose data and research questions that interest you. The project description includes an example project using Trump tweets and FRED-MD data; if you have trouble coming up with your own idea for the project, you are welcome to implement a version of this project.

Details on Project Components

Your project grade will be composed of the following. For all Gradescope submissions below, only one submission per group is needed, but you must add all group members to the submission.

10% **Project Plan.** By **Wednesday May 20th at 11:59 pm**, submit a 1 page project outline on Gradescope. This should include a rough plan for the project, a few summary statistics from data collected for the project, and the results from at least one linear regression on the data. To receive full points, the group must also have signed up for a presentation slot before Wednesday May 20th.

50% **Paper.** A write-up of minimum **4 pages** and maximum **8 pages**, describing concisely and clearly your dataset, the models you used to analyze the dataset, and a summary of your results. This is due on Gradescope on **Tuesday, May 26th at 11:59 pm**. The writeup must include an appendix explaining each group member's contribution.

10% **Code.** A zip file of the code used to generate the results for your project, due on Gradescope on **Tuesday, May 26th at 11:59 pm**. Do not submit data with your code.

30% **Presentation.** A 10 minute presentation to Prof. Munro on **Wednesday, May 27th from 1pm - 5pm**: roughly 5 minutes of presentation, followed by Q&A about your results and code. In the case of final exam conflicts, at most 2 group members may be absent from the presentation. Sign-up for presentations will be via a Google sheet distributed via Canvas.

Honor Code

- Using AI tools to scrape and clean data, to generate features, to brainstorm and design your project, and to write code for your project is acceptable and encouraged. However, you must be able to explain what each line of code does, and you should not use AI to generate your written report.
- Do not use a project that you have written for another class.
- Data scraping is often not viable for large websites like Zillow, Redfin, Amazon, etc; it may violate their Terms of Service and can lead to account or IP bans, and in some rare cases, legal action. If you scrape, please be careful, do not send too many requests too quickly, and avoid violating Terms of Service. Given the limited time available for the project, using pre-existing datasets or APIs may be preferred.

Useful Resources

- Trump Tweets Data [Link] and FRED-MD Data [Link]
- Amazon Reviews Dataset [Link]
- Airbnb Dataset (e.g. for Chicago) [Link]
- HuggingFace text models [Link]
- Polymarket API [Link] and Kalshi API [Link]
- UChicago Libraries data guide [Link]
- Reddit API [Link]
- TabFPN for tabular data: [Link]

Suggestions

- Always run analysis on a very small and clean subset of your dataset before trying to produce results on a larger dataset. For the final version, I suggest using a dataset of moderate size (e.g. at most a few hundred MB for text and numerical data).
- Cleaning and preparing data takes more time than you might expect. You should collect data and run some simple models from the beginning of the class well in advance of the May 20th deadline for the project outline.
- To use AI tools effectively, break a large task into many small tasks, and get help with one small task at a time. Verify the correctness of each subtask before moving on to the next one.